1 Modular Arithmetic Tools

1.1 Divisibility in $\mathbb{Z}$

1.1 Let $a, b \in \mathbb{Z}$ be integers such that $\text{mcd}(a, b) = 1$. Show that the greatest common divisor of $a + b$ and $a - b$ is either 1 or 2.

1.2 Show that, for every $a, b, m \in \mathbb{Z}$, $\text{mcd}(ma, mb) = m \cdot \text{mcd}(a, b)$.

1.3 Is it true that, for $a, b \in \mathbb{Z}$, $\text{mcd}(3a - b, b - 2a) = \text{mcd}(a, b)$? Explain why.

1.4 Compute $\text{mcd}(28n - 5, 35n - 8)$, for $n \geqslant 1$.

1.5 Find the greatest common divisor and the Bézout coefficients for each of the following pairs of integers (check its validity):

a) $1492, 1776$;

b) $7469, 2464$;

c) $500, 11312$.

1.6 Compute the least common multiple of each pair of integers from the previous exercise.

1.7 Show that the fraction $\frac{4n+5}{2n+3}$ is irreducible for every $n \geqslant 1$.

1.8 Solve the following Diophantine equations:

a) $93x - 81y = 3$;

b) $71x - 50y = 1$;

c) $42x + 18y = 34$;

d) $283x - 1722y = 31$.

1.9 a) Using the Euclidean algorithm, compute the greatest common divisor $\text{mcd}(1911, 65)$. b) Justify whether the Diophantine equation $3822x + 65y = 26$ has solution or not, in the positive case, describe them all (Indication: you can use the previous item).

1.10 Ask somebody to multiply his/her day of birthday by 12, multiply his/her month of birthday by 31, add the two results, and tell you only this final sum. Can you guess, just with this information, when is his/her birthday?

1.11 Show that if ab is a perfect square and $\text{mcd}(m, n) = 1$, then a and b are both perfect squares as well.

1.12 Show that if a and b are integers and $a^2 | b^2$, then $a | b$.

1.13 Find the smallest natural number n such that $n/2$ is a perfect square, $n/3$ is a perfect cube, and $n/5$ is a perfect fifth power.

1.14 Find the decomposition of the six integers involved in Exercise 1.5, as product of primes. Check the formulas relating these decompositions with their respective greatest common divisor and least common multiple (computed before).

1.2 Congruences

1.15 Do the following calculations in the indicated modular arithmetic:

a) $8 \cdot (7^2 - 2) + 6$ a $\mathbb{Z}/12\mathbb{Z}$;

b) $(8^3 - 1)^4$ a $\mathbb{Z}/6\mathbb{Z}$;

c) $((4^3)^5 - 1)^4 + 2^4 \cdot (3^4 - 1)$ a $\mathbb{Z}/10\mathbb{Z}$.

1.16 Prove that if $a \equiv b \pmod{m}$ then, for every value of k, $ka \equiv kb \pmod{m}$. Is it true that if there exists k such that $ka \equiv kb \pmod{m}$, then $a \equiv b \pmod{m}$? And what if k is a prime number?

1.17 Prove that if $a \equiv b \pmod{m}$ then, for every value of n, $a^n \equiv b^n \pmod{m}$. Is it true that if there exist n such that $a^n \equiv b^n \pmod{m}$, then $a \equiv b \pmod{m}$?

1.18 Prove that every even prime p which can be written as the sum of two squares must be $p \equiv 1 \pmod{4}$.

1.19 Make a complete list of the invertible elements modulo 2, 3, 4, 5, 6, 7, 8, 9, and 10.

1.20 Are the elements 29, 234, and 512 invertible modulo 1170? In the affirmative case find their inverses.

1.21 Find all the solutions to the following equations:

a) $31x = 1$ in $\mathbb{Z}/101\mathbb{Z}$;

b) $4x = 0$ in $\mathbb{Z}/18\mathbb{Z}$;

c) $6x = 0$ in $\mathbb{Z}/30\mathbb{Z}$.

1.22 Solve the following systems of linear equations over the indicated rings (try different methods for each one, and check the results agree):

a) $\begin{cases} x + 3y = 15 \\ -2x + 14y = 1 \end{cases}$ in $\mathbb{Z}/17\mathbb{Z}$;

b) $\begin{cases} 3x + 2y = 0 \\ 2x - 4y = 2 \end{cases}$ in $\mathbb{Z}/5\mathbb{Z}$ and $\mathbb{Z}/25\mathbb{Z}$;

c) $\begin{cases} 3x + 2y = 0 \\ 4x - 4y = 2 \end{cases}$ in $\mathbb{Z}/5\mathbb{Z}$ and $\mathbb{Z}/25\mathbb{Z}$;

d) $\begin{cases} 3x + 2y + z = 0 \\ x - y - 4z = 2 \\ -x + 2y + 3z = 3 \end{cases}$ in $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/7\mathbb{Z}$.

1.23 Decide which of the following matrices are invertible over the indicated rings; in case they are, find the inverse:

a) $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ in $\mathbb{Z}/3\mathbb{Z}$.

b) $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ in $\mathbb{Z}/10\mathbb{Z}$.

c) $\begin{pmatrix} 3 & 2 & 1 \\ 1 & -1 & -4 \\ -1 & 2 & 3 \end{pmatrix}$ in $\mathbb{Z}/3\mathbb{Z}$, $\mathbb{Z}/5\mathbb{Z}$ and $\mathbb{Z}/10\mathbb{Z}$.

d) $\begin{pmatrix} 1 & 2 & 0 & 0 \\ 3 & -4 & 0 & 1 \\ 0 & 0 & 2 & -1 \\ 0 & 3 & 2 & 5 \end{pmatrix}$ in $\mathbb{Z}/2\mathbb{Z}$, $\mathbb{Z}/3\mathbb{Z}$, $\mathbb{Z}/5\mathbb{Z}$, $\mathbb{Z}/7\mathbb{Z}$, $\mathbb{Z}/9\mathbb{Z}$, and $\mathbb{Z}/55\mathbb{Z}$.

1.3 The Chinese reminder theorem

1.24 (Sun Tsu Suan-Ching, s. IV) We have a collection of objects; when counted in groups of three there are two remaining, when counted in groups of five there are three remaining, and when counted in groups of seven there are two remaining. How many objects do we have?

1.25 Solve the following systems of congruences:

$$a)\ \begin{cases} x \equiv 10 \pmod 7, \\ x \equiv 6 \pmod 9, \\ x \equiv 1 \pmod{10}. \end{cases} \qquad b)\ \begin{cases} x \equiv 3 \pmod 4, \\ x \equiv 1 \pmod{15}, \\ x \equiv 2 \pmod 7. \end{cases} \qquad c)\ \begin{cases} 3x \equiv 2 \pmod 4, \\ 4x \equiv 7 \pmod{15}, \\ 8x \equiv -1 \pmod 7. \end{cases}$$

1.26 Let $m_1, \ldots, m_n > 1$ be integers coprime to each other, and let $M = m_1 \cdots m_n$. Prove that, for every $a, b \in \mathbb{Z}$, $a \equiv b \pmod M$ if and only if $a \equiv b \pmod{m_i}$, for every $i = 1, \ldots, n$. Is there any relation with the Chinese Remainder Theorem?

1.27 Check, by hand, whether the maps

$$\begin{array}{rcl} f\colon \mathbb{Z}/18\mathbb{Z} & \to & \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/9\mathbb{Z}; \\ [x]_{18} & \mapsto & ([x]_2, [x]_6, [x]_9) \end{array} \qquad \begin{array}{rcl} f\colon \mathbb{Z}/18\mathbb{Z} & \to & \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/9\mathbb{Z}. \\ [x]_{18} & \mapsto & ([x]_2, [x]_9) \end{array}$$

are bijective or not. Does it match with a theorem you know about this?

1.4 Fermat and Euler's theorems

1.28 a) Find the inverse of 37 in $\mathbb{Z}/98\mathbb{Z}$.
b) Compute the value of $37^{55} \cdot 53^{13} \pmod{98}$.

1.29 Compute the values of (a) $513^{1017} \pmod 6$; (b) $2^{154} \pmod 5$; (c) $3^{178} \pmod{12}$.

1.30 Compute the last two digits of the huge number 103^{1243}.

1.31 Find all the numbers n for which $10^n + 3$ is multiple of 7.

1.32 Find all the numbers n for which $4^n + 5$ is multiple of 15.

1.33 a) Solve the equation $2^{2021}(x - 2^{2022}) = 4^{2023} \pmod{13}$.
b) Deduce from it a description of all solutions to the equation $2^{2021}(10 \cdot 3^n - 2^{2022}) = 4^{2023} \pmod{13}$.

1.34 Compute $2022^{2022^{2022}} \pmod{19}$ and $\pmod{21}$.

1.35 Compute the following values of the Euler phi function: $\phi(128)$, $\phi(1123)$, $\phi(1125)$, $\phi(29270241)$, and $\phi(12648925575)$.

1.36 Order the following rings by decreasing amount of invertible elements: $\mathbb{Z}$, $\mathbb{Z}/11\mathbb{Z}$, $\mathbb{Z}/12\mathbb{Z}$, $\mathbb{Z}/18\mathbb{Z}$, and $\mathbb{Q}$.

1.37 The Mersenne numbers. The q-th Mersenne number is defined as $M_q = 2^q - 1$. These numbers are not always prime, but were found and studied by the french mathematician Marin Mersenne as an important source of prime numbers.

a) show that if M_q is prime then q is prime as well (hint: $x^n - 1 = (x-1)(x^{n-1} + x^{n-2} + \cdots + x + 1)$);

b) show that the converse is not true, by giving the smallest possible counterexample;

c) in the cases where q is primer and M_q is not, the primes dividing M_q are very limited: show that if p, q are primes and $p | M_q$ then $p \equiv 1 \pmod q$ (hint: use the previous item);

d) Decide whether M_{13} or M_{17} are primes or not (you may use calculator or computer, but it is not about doing a big number of divisions: use the previous items to drastically reduce the computation).

1.5 True/false problems

1.38 Decide whether the following assertions are true or false, justifying conveniently your answers with an argument or a counterexample:

a) In the ring $M_2(\mathbb{Z})$ there are two elements $A, B \neq 0$ such that $AB = 0$.

b) In the ring $M_2(\mathbb{Q})$ it is true that $AB = AC \Rightarrow B = C$.

c) There is a ring containing elements opposed to themselves.

d) In a ring R, if two elements $a, b \in R$ are invertible, then so is ab and $(ab)^{-1} = a^{-1}b^{-1}$.

e) There is a ring R containing two elements $a, b \in R$ such that $a^2 = 1$, $b^2 = 1$, but $(ab)^2 \neq 1$.

f) In $\mathbb{Z}$, $\operatorname{mcd}(249018, 34650) = 462$.

g) In $\mathbb{Z}$, if $a \neq 2$, $a|(b+c)$ and $a|(b-c)$, then $a|b$ and $a|c$.

h) In $\mathbb{Z}$, if a és odd, $a|(b+c)$ and $a|(b-c)$, then $a|b$ and $a|c$.

i) In $\mathbb{Z}$, if $d|a$, $d|b$ and we write $a = da'$ and $b = db'$, then a' and b' are coprimer.

j) The Diophantine equation $13x + 23y = 4$ has no integer solutions.

k) All the integral solutions to the equation $2x - 6y = 8$ are of the form $x = 16 + 3\lambda$, $y = 4 + \lambda$, $\lambda \in \mathbb{Z}$.

l) In $\mathbb{Z}$, $\operatorname{mcd}(a, \operatorname{mcm}(b, c)) = \operatorname{mcm}(\operatorname{mcd}(a, b), \operatorname{mcd}(a, c))$.

m) Less than a fourth of the numbers between $1, 2, \ldots, 100$ are primers.

n) The integer 2347 is prime.

o) If $[a]_m \cap [b]_m \neq \emptyset$ then $m|(a-b)$.

p) For every $a \in \mathbb{Z}$ and $m \geqslant 2$, we have that $[a]_{2m} \subseteq [a]_m$, but they are not equal.

q) If $[a]_m \cap [b]_{2m} \neq \emptyset$ then $[a]_{2m} = [b]_{2m}$.

r) In $\mathbb{Z}/6\mathbb{Z}$ we have that $3 = 0$ because "$2 \cdot 2 = 2 \cdot (-1) \Rightarrow 2 = -1 \Rightarrow 3 = 0$".

s) The equation $2x = 7 \pmod{10}$ has no solution, while the equation $2x = 6 \pmod{10}$ has two.

t) The system of equations $\left.\begin{array}{c} 2x + 3y = 3 \\ x + 2y = 4 \\ 5x - 4y = 2 \end{array}\right\}$ is compatible and determined in $\mathbb{Z}/13\mathbb{Z}$.

u) If $a \equiv b \pmod{r}$ and $c \equiv d \pmod{s}$, then $a + b \equiv c + d \pmod{t}$ where $t = \operatorname{mcm}(r, s)$.

v) If $a \equiv b \pmod{r}$ and $c \equiv d \pmod{s}$, then $ab \equiv cd \pmod{t}$ where $t = \operatorname{mcd}(r, s)$.

w) $11^{11111} \equiv 6 \pmod{13}$.

x) $22^{2222} \equiv 33^{3333} \pmod{17}$.

y) If $r \equiv s \pmod{p}$, then $a^{r-1} \equiv a^{s-1} \pmod{p}$.

z) If p and q are different primes and $a \in \mathbb{Z}$ satisfies $p \nmid a$ and $q \nmid a$, then $a^{(p-1)(q-1)} \equiv 1 \pmod{pq}$.